Paper Code – CSMG-402-6 **Paper Name – Computational Biology**		**Full Marks: 100**
Module	**Topics**	**Hours**
Module 1: Cell Biology	System Biology, Central Dogma of Molecular Biology, Biological Database: Sequence data, Gene expression data, Micro-array experiment, NCBI database, Challenges faced in the integration of biological data, Data management and data integration in bio-informatics, Introduction to Next Generation Sequencing.	**4**
Module 2: Sequence Analysis	Sequence similarity, homology, and alignment, Pair wise alignment: scoring model, dynamic programming algorithms, heuristic alignment, and pair wise alignment using Hidden Markov models (HMM), Sequence file formats – GenBank, FASTA, etc., Sequence similarity based search engines (BLAST), Multiple alignment: scoring model, local alignment gapped and un-gapped global alignment, Motif finding: motif models, finding occurrence of known sites, discovering new sites, Amino Acid, Protein, Phylogenetic tree construction: Neighbor Joining Algorithm.	**8**
Module 3: Biological Pathways	Gene regulatory network, Transcription factors, Signal Transduction, Protein-Protein interaction, Boolean Network, Stochastic gene networks, Network connectivity.	**4**
Module 4: Clustering, Classification and Rule mining	Clustering algoritms: k-means, k-medoid, Isodata, AGNES, DIANA, BIRCH, DBSCAN, Fuzzy C-Means, Cluster validity indices: DB-Index, Dunn Index, Xie-Beni Index etc. Classification algorithms: Bayes' theorem, Naïve Bayes' classifier, Bayesian belief network, Decision Tree. Association rule Mining: Apriori, FP-Growth.	**14**
Module 5: Statistical approach	Information theory (Entropy), Prediction using linear regression, multiple regression, predicting reading frames, maximal dependence decomposition, Expectation-maximization, Bayesian model, Gaussian Mixture Model(GMM), P-value statistics(GO), z-score, t-test, F-test, Validation parameters: True positive, Sensitivity, Specificity, FDR, Accuracy.	**10**

Textbook:

1. Molecular Biology of the Cell, Bruce Alberts, Alexander Johnson, Julian Lewis, Martin Raff, Keith Roberts, Peter Walter, New York: Garland Science; 2002.
2. Data Mining Concepts and tecniques, Han & Kamber, Elsevier
3. Introduction to Bioinformatics, Arthur M. Lesk, Oxford University Press

References:

1. Bioinformatics: Sequence and Genome Analysis, David W. Mount, Cold Spring Harbor Laboratory Press.
2. Bioinformatics And Functional Genomics: A Short Course, Jonathan Pevsner, Wiley-Liss